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Deep learning based artificial intelligence model for identifying swallow types in esophageal high-resolution manometry

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Abstract

Background: This study aimed to build and evaluate a deep learning, artificial intelligence (AI) model to automatically classify swallow types based on raw data from esophageal high-resolution manometry (HRM).

Methods: HRM studies on patients with no history of esophageal surgery were collected including 1,741 studies with 26,115 swallows labeled by swallow type (normal, hypercontractile, weak-fragmented, failed, premature) by an expert interpreter per the Chicago Classification. The dataset was stratified and split into train/validation/test datasets for model development. Long short-term memory (LSTM), a type of deep-learning AI model, was trained and evaluated. The overall performance and detailed per-swallow-type performance were analyzed. The interpretations of the supine swallows in a single study were further used to generate an overall classification of peristalsis.

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Conflicts of interest:
John E. Pandolfino and Peter J Kahrilas hold shared intellectual property rights and ownership surrounding FLIP panometry systems, methods, and apparatus with Medtronic Inc.
Dustin A. Carlson: Medtronic (Speaking. Consulting).
John E. Pandolfino: Sandhill Scientific (Consulting, Speaking), Takeda (Speaking), Astra Zeneca (Speaking), Medtronic (Speaking. Consulting), Torax (Speaking, Consulting), Ironwood (Consulting), Impleo (Grant).
Wenjun Kou, Galal Osama Galal, Matthew William Klug, Vladislav Mukhin, Mozziyar Etemadi disclose no conflicts of interest.

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Key Results: The LSTM model for swallow type yielded accuracies from the train/validation/test datasets of 0.86/0.81/0.83. The model's interpretation for study-level classification of peristalsis yielded accuracy of 0.88 in the test dataset. Among model misclassification, 535/698 (77%) swallows and 25/35 (71%) studies were to adjacent categories, e.g. normal to weak or normal to ineffective, respectively.

Conclusions & Inferences: A deep learning AI model can automatically and accurately identify the Chicago Classification swallow types and peristalsis classification from raw HRM data. While future work to refine this model and incorporate overall manometric diagnoses are needed, this study demonstrates the role that AI will serve in the interpretation and classification of esophageal HRM studies.

Keywords

high-resolution manometry; esophageal peristalsis; machine learning

INTRODUCTION

High-resolution Manometry (HRM) is the primary method for the clinical evaluation of esophageal motility and the diagnosis of esophageal motility disorders[1]. The current interpretation of HRM relies on feature-based analysis algorithms, which first evaluate the specific landmarks and predefined metrics such as distal latency (DL) and distal contractile integral (DCI) in an algorithmic way, and then derive diagnoses based on the established rules[2]. Specifically, the Chicago Classification (CC) categorizes each HRM study into one of multiple esophageal motility diagnoses based on a decision tree algorithm[1]. Features of each test swallow are used to determine the swallow type (normal, hypercontractile, weak-fragmented, failed, premature) and finally the peristalsis classification is determined based on the established CC algorithm[1,3].

While the feature extraction and classification algorithms provide a framework for uniform esophageal motility interpretation with the CC, manual and subjective input are required. Hence, the interpretation of HRM is subject to inter-rater variability and, consequently, rater associated inaccuracy[4,5]. Furthermore, several landmarks generated by the analytic software are reliant on manual positioning of anatomic and physiologic landmarks; inconsistent landmark identification is a major contributor to inter-rater disagreement. Reliance on subjective experience also poses a challenge in medical training; a recent post-training evaluation survey from GI fellows showed only 84 percent were confident in the interpretation of esophageal manometry[6].

Machine learning could potentially alleviate some of the above issues associated with the clinical interpretation of HRM interpretation based on its proven success in other spheres of medicine[7]. Machine learning, in a broad sense, can be classified into two types. One type is based on processed low-dimensional features/outcomes (such as DL and DCI) and focuses on the improvement of classification. The other approach is directly based on large dimensional raw data (i.e. pressure measurements) to assign the classification label with minimal manual 'pre-processing'. The later approach is often referred to as deep learning, due to its complexity in modeling structure and its reliance on large datasets[8].

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The deep learning methodology was adopted in this work to directly classify swallow type based on raw HRM data. Note that machine learning has already found wide application in various fields of medicine, including gastroenterology[9–11]. However, the major focus in gastroenterology has been on the detection of intestinal malignancies or premalignant lesions from endoscopy videos or x-ray images. Hence, the application of a machine learning model to HRM is novel. Specifically, this work aimed to develop a deep learning AI model to classify swallow type from HRM data and evaluate its accuracy against expert-assigned labels.

METHODS

Subjects

Adult patients (age 18–89) presenting to the Esophageal Center of Northwestern for evaluation of esophageal symptoms between 2015 and 2019 who completed clinical HRM studies with data maintained in an esophageal motility registry were prospectively evaluated. Patients with previous foregut surgery (including previous pneumatic dilation) or technically limited HRM studies were excluded. HRM data from 1,741 consecutively evaluated patients were included. There is some overlap with a previously described cohort[12]. The study protocol was approved by the Northwestern University Institutional Review Board.

HRM protocol and diagnosis label

After a minimum 6-hour fast, HRM studies were completed using a 4.2-mm outer diameter solid-state assembly with 36 circumferential pressure sensors at 1-cm intervals (Medtronic Inc, Shoreview, MN). The HRM assembly was placed trans-nasally and positioned to record from the hypopharynx to the stomach with approximately three intra-gastric pressure sensors. After a 2-minute baseline recording, the HRM protocol was performed with ten, 5-ml liquid swallows in a supine position and five swallows in an upright position. Manometry studies were analyzed using ManoView version 3.0 analysis software (Medtronic) to measure basal EGJ pressure (end-expiration), the integrated relaxation pressure (IRP), DCI, and DL. Then the label of swallow type (normal, weak/fragmented, failed, premature, or hypercontractile) was determined according to the CC rules[1]. These swallow-type labels ("true labels") were assigned by an expert in HRM interpretation and one of the founders of the CC and lead in its successive updates (JEP)[2].

HRM dataset

HRM data were processed and the dataset was restricted to contain HRM studies that satisfied the following criteria: 1) clearly labeled swallow type based on CC rules; 2) having 10 supine and 5 upright swallows with swallow type labels. Based on distribution of study labels, the whole dataset was split to three datasets (training, validation, and test) for training models on identifying the swallow type. To reduce the confusion between similar swallow types, the weak and fragmented groups were merged into a weak-fragmented group which is in line with the recently published CC v4.0. As shown in Table 1, the split was stratified so that: 1) swallows within the same study were in the same dataset to avoid *information leaking*; and 2) the swallow type distribution among training, validation, and testing datasets was statistically similar.

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In addition to labeling of swallow types, both the true-swallow labels and the AI assigned labels of the 10 supine swallows were applied to the criteria defined by the CC v4.0 to generate a study-level classification of peristalsis. These labels were assigned as absent contractility (100% failed swallows), spasm (≥20% premature swallows), hypercontractile (≥20% hypercontractile swallows, but not spasm), ineffective-conclusive (≥50% failed swallows or >70% weak or ineffective swallows), or normal (not meeting preceding criteria).

Data preprocessing and deep learning model

For the deep learning model, the input datapoint is the pressure data from one swallow consisting of recordings of 36 pressure sensors that sample at frequency of 100Hz. To reduce the data dimension with minimal information loss, the swallow data is retained as a 24-second window and down-sampled from 100Hz to 10Hz. Hence, the datapoint assumes the shape of 240•36. Details and justification of this data preparation were previously reported[12]. Neural network models of various types and architecture were first explored and compared leading us to conclude that the long-short time-memory (LSTM) model with attention was the optimal model candidate[8]. The implementation of the LSTM model was based on Tensorflow, a machine learning library in Python[13]. Training was completed on a single K80 machine. Modeling testing and evaluation was conducted with metrics including the overall accuracy as well as precision, recall, and F1-score per swallow type.

Model architecture and training history

The LSTM model includes multilayer neural networks obtained by numerical experiments. The weights of each layer were trained with a large training dataset and evaluated on the validation dataset iteratively, so that separating patterns could be ‘memorized’ and ‘learned’. The training process was finished at about 17 iterations (epochs) at which point the validation accuracy plateaued. Additional details on model design and training are included in the Appendix (Table S1; Figure S1). After training, the model was applied to identify swallow types in the testing dataset and those data were then used to generate an overall study-level classification of peristalsis.

RESULTS

Dataset

The total dataset included 1,741 HRM studies. HRM study diagnoses by CC v3.0 included 1,741 patients: 272 (15.5%) achalasia (66 type I, 129 type II, and 77 type III), 295 (16.9%) EGJ outflow obstruction, 15 (1%) distal esophageal spasm, 40 (2%) hypercontractile, 12 (1%) fragmented peristalsis, 79 (4.5%) absent contractility, 228 (13%) ineffective esophageal motility, and 800 (46%) normal motility.

Model performance and evaluation

Table 2 illustrates performance of the LSTM model on swallow type in the training, validation, and testing datasets. The overall accuracy among the three datasets was similar. Precision, recall, and F1 scores were greatest for normal and failed swallow types and lowest for premature swallows. Sources for agreement and misclassification from the testing dataset are displayed as a confusion matrix in Figure 1. Of 698 misclassified swallows, 77%

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were misclassified by the model into an 'adjacent' swallow type with 304/698 (44%) of misclassifications occurring between weak and normal swallow types and 152/698 (22%) occurring between failed and weak. Examples of misclassified swallows are displayed in Figure 2.

The prediction of overall study-level classification of peristalsis was then derived from the model-assigned swallow types of all the swallows within that study. The overall accuracy in the testing dataset is 0.88 as shown in Table 2. *Normal peristalsis and Absent* contractility had the greatest precision and recall scores. Of 35 misclassified *studies*, 71% were misclassified by the model into an 'adjacent' classification with 13/35 (37%) of misclassifications occurring between normal and ineffective classifications.

DISCUSSION

The major findings of this study were that an automated AI model was able to accurately identify swallow types from raw HRM data and apply that data to generate study-level peristaltic classifications. Because the AI model utilized a deep learning technique that utilized raw pressure data, it was able to automatically classify swallows with minimal operator input. Applied to a large dataset of 1,741 patients and 26,115 swallows, the AI model yielded labeling accuracies of 0.83 and 0.88 for swallow types and study-level peristaltic classifications, respectively. Furthermore, when the model output was 'incorrect', it was often related to adjacent swallow classifications and misclassifications of minimal clinical significance, e.g. label of weak versus normal swallow or ineffective versus normal esophageal motility. Thus, this application of AI to HRM interpretation demonstrates its potential to guide clinical practitioners in the interpretation of HRM.

Artificial intelligence and machine learning are being increasingly utilized to assist interpretation of large complex datasets in medicine. In gastroenterology, the usefulness of AI to assist in interpretation of endoscopic imaging was recently demonstrated for the identification of colon polyps and Barrett's metaplasia[9]. The graphical nature of the topographical pressure display of HRM, or Clouse plots, is therefore amenable to AI techniques in a similar manner[14]. We previously applied another deep learning AI approach (variational auto-encoder) to raw HRM data and demonstrated its potential to extract separating features in an unsupervised manner[12]. The present study, instead, uses a supervised approach to automatically label swallow type from raw HRM data and further derives the study-level label of peristaltic function.

A factor related to AI-model performance relates to the degree of similarity of patterns among certain swallow types, such as normal, weak-fragmented and failed. These swallow types occur along a spectrum of esophageal function and hence the majority (77% swallows and 71% of studies in test cohorts) of the misclassifications were to adjacent swallow or peristalsis classifications (Figures 1 and 3). The 'true' label, and differentiation along this spectrum was determined by applying thresholds of HRM metrics (e.g. DCI) generated by the commercial analytic software. Although these thresholds are based on statistically defined normal values and can impact esophageal bolus clearance, the 'ideal' thresholds remain undefined. For example, the DCI threshold of 450 mmHg•s•cm to differentiate

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normal from weak swallows had an 83% positive agreement with the line tracing assignment of swallows as ineffective[15]. Thus, these thresholds are somewhat arbitrary and carry minimal clinical impact, particularly given the uncertain clinical significance of an HRM classification of ineffective esophageal motility[16]. Further, cases in certain 'borderline' regions could create disagreement even between experts at HRM interpretation. This was evident with fair to good inter-rater agreement among experts classifying HRM studies per the CC (kappa values ranging from 0.24–0.82 between motility classifications)[4]. However, this issue could be alleviated if the AI model is not a deterministic label, but more focused on probabilities of each label. The probability scores (e.g. probabilities for identification of each class of swallow) could also serve as the recommendation to clinicians for manual/expert evaluation. Additional application of the probabilistic outcomes from an AI model is a future research direction.

Detailed evaluation of model performance is necessary to understand the strengths and weaknesses of the AI model, increasing transparency and confidence in usage. The similar accuracies among the train/validate/test datasets indicated there was no apparent over-fitting while the differences in precision/recall/F1-score across classes/labels is likely related to sample imbalance (i.e. support). For example, the performance per class likely depends on sample number of that class in dataset during training and testing, as intuitively the neural network could better memorize/differentiate features if it is fed more samples for a particular class. This phenomenon underpins the importance of a well-curated dataset, not only in its size, but also in its distribution across swallow types. This also suggests the AI model might not be expected to perform equally well in discerning rare swallow types. This was evident with the lowest accuracy of the model observed with the identification of premature swallows and spasm, which were infrequently observed in this cohort of consecutive patients consistent with the clinical rarity of this finding[4]. While the approach utilized in this study has the benefit of retaining the prevalence of esophageal phenotypes seen in the clinical settings, other methods such as applying class weights and/or data oversampling may also be beneficial for future work. Moreover, we hypothesize, that with incorporation of additional training data, potentially even a fortified dataset to augment training for rare cases such as spasm, the model accuracy would be expected to improve. Ultimately, the model described in this work could be applied in the clinical setting to assist in interpretation of HRM study by providing recommendations to a rater.

While this work demonstrates strength in the novel application of a deep learning AI model to automatically interpret raw HRM data with demonstrated accuracy of a large sample of HRM studies, it also has limitations. One limitation is that the model is based on single swallow data, instead of the whole study. However, understanding patterns of swallow-level data is the backbone to interpreting the whole study. Labels for individual swallows obtained from studies completed prior to the consensus development process of CCv4.0 could easily be applied to the updated criteria proposed by CCv4.0[3]. However, a future research direction is to expand upon this architecture to build a study-level AI model that will directly generate the HRM diagnosis with added input of the EGJ outflow assessment related to the series of swallows.

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In conclusion, an automated AI model was able to accurately identify swallow types and study-level peristaltic classification from raw HRM data. When model 'misclassifications' occurred, they likely carried minimal clinical impact. While improvement in model performance is expected with additional training data, future directions also include model development to interpret HRM diagnoses at the study level, as well as to implement a probabilistic model output to enhance its application as a clinical decision support tool. Overall, this study demonstrates that the AI holds great potential to improve HRM interpretation.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data availability:

Research data is not available for sharing.

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True label	**Hypercontractile, n (%)**	**49 (49)**	48 (48)	1 (1)	1 (1)	0
	Normal, n (%)	31 (14)	**2043 (90)**	118 (5)	45 (2)	23 (1)
	Weak, n (%)	0	186 (37)	**217 (44)**	90 (18)	4 (1)
	Failed, n (%)	0	19 (2)	62 (6)	**934 (91)**	7 (1)
	Premature, n (%)	4 (4)	22 (19)	7 (6)	30 (27)	**49 (44)**
		Hyper-contractile	**Normal**	**Weak**	**Failed**	**Premature**
		Model-identified label				

Figure 1. Confusion matrix for the identification of swallow type.

Values reflect the number of swallows and percentage of the true labels from the testing dataset. Dark grey shaded boxed indicate agreement while the lighter shaded boxes indicate a misclassification to an adjacent category (premature was not considered along this spectrum).

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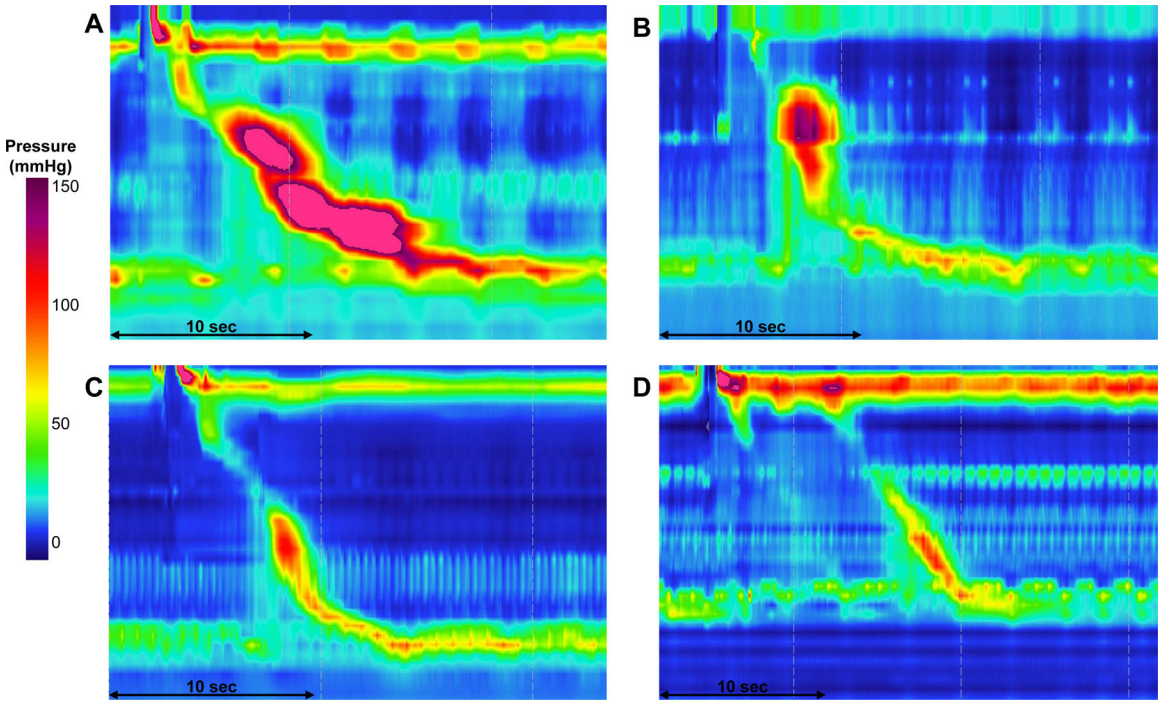

Figure 2. Examples of incorrectly identified swallows.

A) True label was hypercontractile, model-identified label was normal. The distal contractile integral (DCI) was 8,300 mmHg•s•cm, and labeled hypercontractile using the threshold DCI value of 8,000 mmHg•s•cm. The peristalsis propagates in a normal manner without multi-peaked contractions and this is most likely normal despite the true label of hypercontractile. **B**). True label was normal, model-identified label was premature. The distal latency (DL) of this swallow was 4.6 seconds, and thus not labeled premature using a threshold DL value of 4.5 seconds. The contractile front velocity of this swallows was 9 cm/second. **C**) True label was fragmented, the model-identified label was normal. There is a 6 cm peristaltic break in the 20mmHg isobaric contour (threshold value is 5 cm) at the transition zone that led to the fragmented label. The DCI was 1,200 mmHg•s•cm. Thus, in A-C, incorrectly model-identified swallows were related to measurements approximating CC thresholds. **D**) True label failed, model-identified label normal. Primary peristalsis failed after the test swallow, but a normal secondary peristaltic wave then occurred. The DCI of the secondary peristaltic wave was 900 mmHg•s•cm. Figure used with permission from the Esophageal Center of Northwestern.

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True label	**Hypercontractile, n (%)**	**8 (67)**	4 (33)	0	0	0
	Normal, n (%)	3 (2)	**153 (94)**	4 (2)	0	2 (1)
	Ineffective, n (%)	0	9 (21)	**30 (71)**	3 (7)	0
	Absent, n (%)	0	0	2 (5)	**37 (95)**	0
	Spasm, n (%)	1 (9)	2 (18)	5 (45%)	0	**3 (27)**
		Hyper-contractile	**Normal**	**Ineffective**	**Absent**	**Spasm**
		Model-identified label				

Figure 3. Confusion matrix from study-level prediction of peristaltic/contractile classifications.
The individual supine swallow-type predictions were cumulatively applied at the patient-study level to assign a peristaltic/contractile classification[3]. Values reflect the number of patients-studies and percentage based on true label from the testing dataset. Dark grey shaded boxed indicate agreement while the lighter shaded boxes indicate a mis-classification to an adjacent category (spasm was not considered along this spectrum).

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Table 1:

Sample distribution of train/valid/test datasets of swallow types used in this work.

Values reflect the number of each swallow type included in each dataset.

Label details (swallow type)	Train #	Valid #	Test #
Hypercontractile	507	99	99
Normal	10070	2083	2260
Weak-fragmented	2524	564	497
Failed	4677	924	1022
Premature	567	110	112

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Table 2:

Performance of LSTM models on swallow type for the training, validation and testing datasets, as well as overall study-level classification of peristalsis for the testing dataset.

Model	Type	Precision	Recall	F1-score	support
Swallow type: Training dataset	***Overall Accuracy***	**0.86**			
	Hypercontractile	0.84	0.78	0.81	567
	Normal	0.89	0.93	0.91	10070
	Weak-fragmented	0.63	0.49	0.55	2524
	Failed	0.88	0.94	0.91	4647
	Premature	0.84	0.56	0.67	507
Swallow type: Validation dataset	***Overall Accuracy***	***0.81***			
	Hypercontractile	0.67	0.51	0.58	110
	Normal	0.86	0.90	0.88	2083
	Weak-fragmented	0.56	0.45	0.50	564
	Failed	0.84	0.91	0.87	924
	Premature	0.41	0.31	0.36	99
Swallow type: Testing dataset	***Overall Accuracy***	***0.83***			
	Hypercontractile	0.58	0.49	0.54	99
	Normal	0.88	0.90	0.89	2260
	Weak-fragmented	0.54	0.44	0.48	497
	Failed	0.85	0.91	0.88	1022
	Premature	0.59	0.44	0.50	112
Study-level classification of peristalsis for the testing dataset	***Overall Accuracy***	***0.88***			
	Hypercontractile	0.73	0.67	0.70	12
	Normal	0.91	0.94	0.93	162
	Ineffective	0.76	0.71	0.73	41
	Absent	0.93	0.95	0.94	39
	Spasm	0.78	0.58	0.67	12